CSA0611 ARTIFICIAL INTELLIGENCE LAB

PROGRAM 2

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AIM

To implement the **8-Puzzle Problem** using the **Breadth-First Search (BFS)** algorithm in Python to find the shortest path from the initial state to the goal state.

Procedure

1. Start the program.
2. Define the initial state and the goal state of the 8-puzzle.
3. Create a queue and insert the initial state.
4. Remove the first state from the queue.
5. Check whether the current state is the goal state.
6. If the goal state is reached, display the solution and stop.
7. Otherwise, generate all possible moves by moving the blank tile (up, down, left, or right).
8. Add the new valid states to the queue if they have not been visited.
9. Repeat the above steps until the goal state is found.
10. Display the final solution and stop.

Algorithm – 8-Puzzle Problem (Breadth-First Search)

1. Start.
2. Define the initial state and the goal state.
3. Insert the initial state into the queue.
4. Remove the first state from the queue.
5. Check whether it is the goal state.
6. If it is the goal state, display the solution and stop.
7. Otherwise, generate all possible next states by moving the blank tile (up, down, left, or right).

8. Add the new states to the queue if they have not been visited.
9. Repeat Steps 4–8 until the goal state is found or the queue becomes empty.
10. Stop.

Pseudocode – 8-Puzzle Problem

```
BEGIN

Initialize Queue
Insert Initial State into Queue
Mark Initial State as Visited

WHILE Queue is not Empty DO

  Remove First State from Queue

  IF Current State = Goal State THEN
    Display Solution
    STOP
  ENDIF

  Generate all Valid Moves
  FOR each New State DO
    IF New State is not Visited THEN
      Add New State to Queue
      Mark New State as Visited
    ENDIF
  ENDFOR
```

ENDWHILE

Display "No Solution Found"

END

Observation

The BFS algorithm explores all possible states level by level and finds the shortest sequence of moves to reach the goal state. The puzzle is solved successfully when the blank tile is moved to the correct position.

Result

The **8-Puzzle Problem** was successfully implemented in Python using the **Breadth-First Search (BFS)** algorithm, and the goal state was reached successfully.